

Bob's Donut Cafe | Introduction to the one-sample t-test

Bob's Donuts

BOB'S SIGNATURE GLAZED DONUT
only 10g added sugar
on average. Yummy but healthy!

but really 10g?
My recipes use 12-20g but not as good!

You/Me (competitor)

The Plan

🍩 Buy a donut on ten random days.

🔪 Lab-test sugar.

The Results

sugar <- c(7.1, 8.5, 9.5, 10.4, 11.2, 11.8, 12.5, 13.3, 14.2, 15.4)

mean(sugar)

#> 11.39

R

ggplot(tibble(sugar), aes(sugar)) +
 geom_dotplot(binwidth = .5, colour = "#7a4a22") +
 geom_vline(xintercept = 10, colour = "#d4af33") +
 geom_vline(xintercept = mean(sugar), colour = "#b5451f",
 linetype = "dashed")

ggplot2

810121416

grams sugar

Gold line: claim (10g). Red dashed: actual average (11.39g).

hmmmm... is this enough evidence that Bob is cheating?

Two possibilities:

- Recipe really is 10g, 11.39g is just natural variation.
- Recipe really is above 10g.

Which one? Let's settle it with a formal hypothesis test.

Hypothesis test framework

State a null hypothesis (H_0) and an alternative hypothesis (H_1).

$H_0 : \mu = 10$

$H_1 : \mu > 10$

true mean sugar exceeds the claim.

ONE-SIDED (WE USE FIRST)

TWO-SIDED (THE DEFAULT)

$H_1 : \mu \neq 10$

true mean sugar differs from the claim in either direction.

Quick check: name the hypotheses

A drug company claims its pill lowers blood pressure. What is H_0 ? ✓ B

A. The pill lowers blood pressure

B. The pill has no effect

C. The pill raises blood pressure

A factory's bolts should be 5cm. You suspect the machine has drifted out of calibration and want to test this. Which H_1 ? ✓ C

A. $\mu > 5$

B. $\mu < 5$

C. $\mu \neq 5$

Measuring the gap: Is 1.39g a lot?

By itself, 1.39g is unclear. We need another scale, one that accounts for how much the sugar content naturally varies among donuts, and how many donuts we weighed.

Variance: how scattered?

High variance

Low variance

less surprising

surprising

Sample size: how much data?

2 donuts

100s of donuts

1.39g diff doesn't mean much

might point to real difference

For the same average gap, the evidence for a significant difference is stronger when the data are less ____ (variable / consistent) and the sample size is ____ (larger / smaller). ✓ variable • larger

Step 1: standard deviation, s

s measures how much individual donuts vary. Average distance from the mean.

$$s = \sqrt{\frac{\sum_i (x_i - \bar{x})^2}{n - 1}} = 2.59 \text{ g}$$

mean 11.39

810121416

Step 2: standard error, SE

Means wobble less than single donuts. Divide s by $\sqrt{n}$.

$$SE = \frac{s}{\sqrt{n}} = \frac{2.59}{\sqrt{10}} = 0.82 \text{ g}$$

Step 3: t statistic

Standardised gap: how many SEs above the claim?

$$t = \frac{\bar{x} - \mu_0}{SE} = \frac{1.39}{0.82} = 1.70$$

s <- sd(sugar)

se <- s / sqrt(length(sugar))

t <- (mean(sugar) - 10) / se

#> t = 1.70

R

Step 4: p -value

Compare t to a t curve with $df = n - 1 = 9$. One-sided $H_1 : \mu > 10$, so p = the upper tail only.

-4-2024

t (standard errors from the claim)

$t = 1.70$

pt(t, df = 9,
 lower.tail = FALSE)

#> 0.062

R

$t = 1.70$, upper-tail $p = 0.062$.

All-in-one in R

t.test(sugar, mu = 10, alternative = "greater")

One Sample t-test

data: sugar

t = 1.6972, df = 9, p-value = 0.06195

alternative hypothesis: true mean is greater than 10

#> ... (CI + estimates clipped) ...

One-sided $p = 0.062 > 0.05$: suggestive, but not quite enough to reject H_0 at $n = 10$.

Practice reading one-sample t-tests, then extend Bob's audit

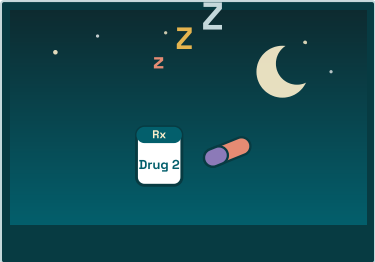
The built-in datasets `sleep` data record extra hours of sleep after two drugs. Here we test whether drug 2 adds more than 0 hours on average.

R

```
data(sleep, package = "datasets")
drug2 <- subset(sleep, group == 2)$extra
drug2
# [1] 1.9 0.8 1.1 0.1 -0.1 4.4 5.5
1.6 4.6 3.4

t.test(drug2, mu = 0,
       alternative = "greater")
```

One Sample t-test
data: drug2
t = 3.6799, df = 9, p-value = 0.002538
alternative hypothesis: true mean is greater than 0
95 percent confidence interval:
1.169334 Inf
sample estimates:
mean of x
2.33



1. What is the null hypothesis being tested?

A. $H_0 : \mu = 0$

B. $H_0 : \mu > 0$

C. $H_0 : \mu = 2.33$

2. What is the sample mean extra sleep?

A. 0

B. 2.33

C. 3.6799

3. How many observations were tested?

A. 9

B. 2

C. 10

4. Match each term to its value, then to its plain-English meaning. Some values are rounded.

Values: a. 9 · b. 0.002538 · c. [1.17, ∞) · d. 3.68

TERM	VALUE	MEANING
t	d. 3.68	b. ≈ # of SDs sample mean is above 0
df	a. 9	a. sample size minus one (n - 1)
p-value	b. 0.002538	d. P(t this extreme H_0 true)
95% CI	c. [1.17, ∞)	c. plausible range for the true mean μ

Meanings: a. n - 1 · b. ≈ # of SDs the sample mean sits above 0 · c. plausible range for μ · d. P(t this extreme | H_0)

5. At $\alpha = 0.05$, what is the decision?

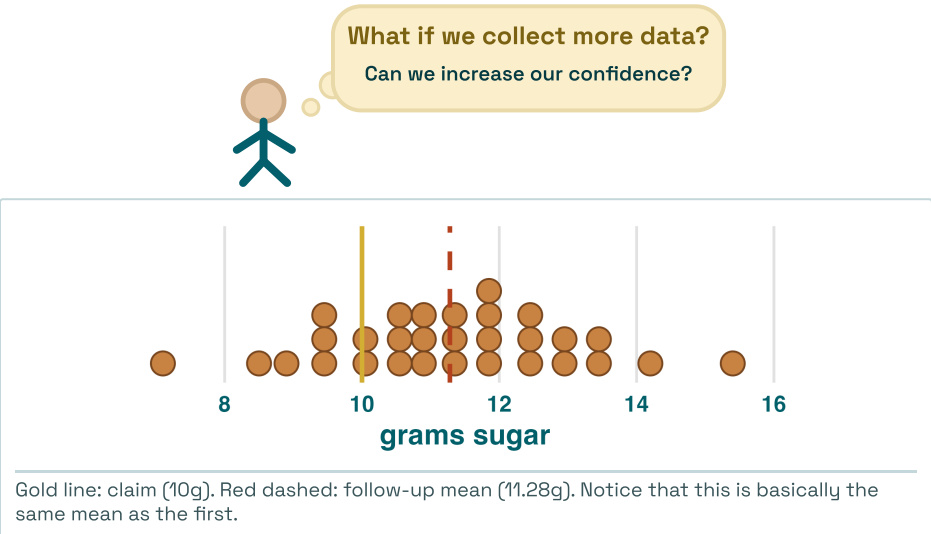
A. Fail to reject H_0

B. Reject H_0

C. H_0 is proven true

Back to the audit of Bob's donuts: More data

The first donut audit had $\bar{x} = 11.39$ and one-sided $p = 0.062$. Close, but not below $\alpha = 0.05$.



```
sugar30 <- c(7.1, 8.5, 9.5, 10.4, 11.2, 11.8, 12.5, 13.3, 14.2, 15.4, 10.9, 9.6, 11.7, 12.3, 10.2, 11.0, 13.1, 9.9, 12.6, 10.7, 11.5, 10.4, 12.0, 9.3, 11.9, 13.6, 10.8, 11.3, 12.8, 8.9)
```

```
t.test(sugar30, mu = 10,
       alternative = "greater")
```

One Sample t-test
data: sugar30
t = 3.9266, df = 29, p-value = 0.000244
alternative hypothesis: true mean is greater than 10
95 percent confidence interval:
10.72612 Inf
sample estimates:
mean of x
11.28

Manual calculations

$s = 1.79 \text{ g}$

$SE = \frac{s}{\sqrt{n}} = \frac{1.79}{\sqrt{30}} = 0.33$

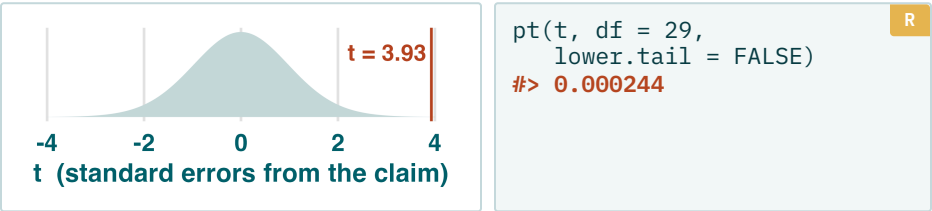
$t = \frac{11.28 - 10}{0.33} = 3.93$

```
s <- sd(sugar30)
#> 1.79

se <- s / sqrt(30)
#> 0.33

t <- (mean(sugar30) - 10) / se
#> 3.93
```

Convert t to p



Interpret the follow-up test

6. What stayed the same from the first donut test?

A. $H_0 : \mu = 10$ and $H_1 : \mu > 10$

B. The sample size

C. The p-value

7. Why is the evidence now stronger?

A. The claim changed

B. The sample mean became significantly larger

C. More data made the SE smaller

8. At $\alpha = 0.05$, what is the decision now?

A. Fail to reject H_0

B. Reject H_0

C. Prove H_0 false

9. In context, the follow-up audit gives strong evidence that Bob's true mean added sugar is ____ 10g. (above / below / equal to)

above

10. Which interpretation is most careful?

A. Every donut has more than 10g of sugar

B. Bob is probably adding 11.28g of sugar to all his donuts

C. The data are unlikely if Bob is truly adding 10g sugar on average